PREP 2 · LESSONS 3–4

From plum pudding to the nucleus

The plum pudding model · alpha scattering · the nuclear model · energy levels

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Describe the structure of an atom. DN 224 [1]

2 Define isotopes. DN 242 [1]

3 State how to calculate the number of neutrons in an atom. DN 716 [1]

Key ideas LESSONS 3–4

Show your working

4 Describe the plum pudding model of the atom. [1]

5 In the alpha scattering experiment, alpha particles were fired at thin gold foil.

a Most alpha particles passed straight through. What does this show? [1]

b A very few bounced back. Explain what this shows about the nucleus. [2]

6 In the Bohr model, electrons orbit the nucleus at **any distance** | **specific distances** [1]

7 An electron moves to a higher energy level when the atom **absorbs** | **emits** electromagnetic radiation. [1]

8 The evidence for the neutron came from **Niels Bohr** | **James Chadwick** | **Ernest Rutherford** [1]